# Chapter 4 homework

```
#|message: false

library(tidyverse)
library(broom)
library(here)
```

```
fitness <- read_csv(here("data", "fitness.csv"))
```

## Homework questions

### 1. Sampling distributions and statistical surprise

Write brief answers.

- What is a sampling distribution of a statistic (like a mean), and what is it useful for? The sampling distribution of a statistic is a theoretical quantity that represents what would happen if you took a samples from a population with a set size, mean, and standard deviation. It is useful for testing the null hypothesis.
- How can a sampling distribution be used to evaluate whether an observed value is "surprising" under a null hypothesis? A sampling distribution can be used to evaluate whether an observed value is "surprising" under a null hypothesis by creating a null distribution, which can be used to find critical PRE values to compare to the observed PRE value. If the observed PRE value is vastly different from the null PRE distribution, then the observed value would be considered "surprising."

### 2. Model comparison test: is the mean resting pulse 72?

Using the Fitness data, evaluate whether the sample of participants is consistent with a population mean of `RSTPULSE = 72` using a **model comparison**.

### 2a. Fit Model C and Model A

Use these models:

- **Model C (null):** the outcome is fixed at 72 with no free parameters

$$Y_i = 72 + \varepsilon_i.$$

- **Model A (mean model):** estimate the mean from the data

$$Y_i = b_0 + \varepsilon_i.$$

```r
fitness <- fitness |>
  mutate(rstpulse_dev = RSTPULSE - 72)

model_c <- lm(rstpulse_dev ~ 0, data = fitness)
model_a <- lm(rstpulse_dev ~ 1, data = fitness)

summary(model_c)
```

```
Call:
lm(formula = rstpulse_dev ~ 0, data = fitness)

Residuals:
   Min     1Q Median     3Q    Max 
 -32.0  -24.0  -20.0  -13.5    4.0 

No Coefficients

Residual standard error: 20 on 31 degrees of freedom
```

```r
summary(model_a)
```

```
Call:
lm(formula = rstpulse_dev ~ 1, data = fitness)

Residuals:
    Min      1Q  Median      3Q     Max 
-13.742  -5.742  -1.742   4.758  22.258 
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -18.26       1.49  -12.26 3.29e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.294 on 30 degrees of freedom
```

### 2b. Compute PRE and F

Compute:

$$\text{SSE}(\cdot) = \sum_i e_i^2$$

(use `deviance()` for an `lm` object)

$$\text{PRE} = \frac{\text{SSE}(C) - \text{SSE}(A)}{\text{SSE}(C)}$$

$$F^* = \frac{\text{PRE}/(P_A - P_C)}{(1 - \text{PRE})/(n - P_A)}$$

```r
sse_c <- deviance(model_c)
sse_a <- deviance (model_a)

pre <- (sse_c - sse_a) / sse_c
pre
```

```
[1] 0.8335267
```

```r
n <- fitness |>
  nrow()

pc <- 0
pa <- 1

fstat <- (pre / (pa - pc)) / ((1 - pre) / ( n - 1))
```

**2c. Identify "surprising" values from the book tables and/or by using `qf()`.**

Fill in the quantities the tables depend on:

- number of participants: n = 31
- number of parameters in Model C: $P_C = 0$
- number of parameters in Model A: $P_A = 1$
- new parameters: $P_A - P_C = 1$
- unused parameters (residual df for Model A): $n - P_A = 30$

Then, using the book tables and/or `qf()`:

- critical PRE at $\alpha = .05$: PREcrit = 0.122
- critical F at $\alpha = .05$: Fcrit = 4.171

```
df1 <- 1
df2 <- 30
alpha <- .05

# PRE
calc_null_pre <- function() {

  null_data <- tibble(rstpulse_dev = rnorm(df2, 0, 72))

  m0 <- lm(rstpulse_dev ~ 0, data = null_data)
  m1 <- lm(rstpulse_dev ~ 1, data = null_data)

  nullpre <- (deviance(m0) - deviance(m1)) / deviance(m0)

  return(nullpre)
}

null_sims <- tibble(sim_number = 1:1000) |>
  rowwise() |>
  mutate(pre = calc_null_pre())

quantile(null_sims$pre, c(.95, .99, 1))
```

```
      95%       99%      100%
0.1177732 0.2035586 0.3673728
```

```r
# F
f_crit <- qf(1 - alpha, df1, df2)
f_crit
```

```
[1] 4.170877
```

### 2d. Conclusions

- Statistical conclusion: Do you reject or retain the null hypothesis at $\alpha = .05$? Explain using the logic of "surprising under the null." You reject the null hypothesis at $\alpha = .05$ because the observed F statistic of 150.209 is vastly different than the F* of 4.17, meaning that the observed F statistic is surprising under the null distribution.
- Substantive conclusion: Give a one-sentence interpretation in plain language. The mean resting pulse of the sample is not 72.

## 3. Alternative F formula using SSR and SSE(A)

Let

$$\mathrm{SSR} = \mathrm{SSE}(C) - \mathrm{SSE}(A)$$

$$F^* = \frac{\mathrm{SSR}/(P_A - P_C)}{\mathrm{SSE}(A)/(n - P_A)}$$

### 3a. Compute F using SSR and SSE(A)

```r
ssr <- sse_c - sse_a
f_stat_ssr <- (ssr/ (pa - pc)) / (sse_a / (n - pa))
```

### 3b–3e. Interpretation

Write brief answers.

- What does the numerator $\text{SSR}/(P_A - P_C)$ represent? The numerator represents mean squares reduced.
- What does the denominator $\text{SSE}(A)/(n - P_A)$ represent? The denominator represents the mean square error.
- What does $F^*$ represent conceptually? F represents the proportional reduction in error per added parameter compared to the proportion of error that remains.
- Compute $t$ for this $F^*$ (use the relationship between $t$ and $F$ when the numerator df is 1).

```
t_stat <- sqrt(f_stat_ssr)
```

## 4. Does running increase pulse rate?

Using the Fitness dataset, compare `RSTPULSE` (resting pulse) to `RUNPULSE` (post-run pulse).

### 4a. Set up Model C and Model A

Use a model comparison where the null corresponds to **no mean change** from resting to post-run.

One convenient approach:

1. Create a difference score $D_i = \text{RUNPULSE}_i - \text{RSTPULSE}_i$.
2. Compare:

    - **Model C:** $D_i = 0 + \varepsilon_i$ (no free parameters)

    $$D_i = 0 + \varepsilon_i.$$

    - **Model A:** $D_i = b_0 + \varepsilon_i$ (estimate the mean difference)

    $$D_i = b_0 + \varepsilon_i.$$

```
fitness <- fitness |>
  mutate(d = RUNPULSE - RSTPULSE)

modelc <- lm(d ~ 0, data = fitness)
modela <- lm(d ~ 1, data = fitness)
```

```r
summary(modelc)
```

```
Call:
lm(formula = d ~ 0, data = fitness)

Residuals:
   Min     1Q Median     3Q    Max 
  92.0  111.0  116.0  123.5  136.0 

No Coefficients

Residual standard error: 116.4 on 31 degrees of freedom
```

```r
summary(modela)
```

```
Call:
lm(formula = d ~ 1, data = fitness)

Residuals:
     Min       1Q   Median       3Q      Max 
-23.9032  -4.9032   0.0968   7.5968  20.0968 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  115.903      1.966   58.95   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.95 on 30 degrees of freedom
```

#### 4b. Compute PRE and F, then compare to critical values

```r
ssec <- deviance(modelc)
ssea <- deviance(modela)
pre_4 <- (ssec - ssea) / ssec
f_4 <- (pre_4 / (pa - pc)) / ((1 - pre_4) / ( n - 1))
f_4
```

```
[1] 3475.443
```

From the book tables at $\alpha = .05$:

- PREcrit = .122
- Fcrit = 4.17

### 4c. Conclusions

- Statistical conclusion (reject/retain the null). I reject the null hypothesis because the observed F statistic of 3475.443 exceeds the F* of 4.17.
- Substantive conclusion (plain-language interpretation). There is some mean change between resting and running pulse.

## 5. With your own data

Return to your own dataset and the model comparison you set up previously. In this section, please use

### 5a. Fit your models

```
gss2024 <- readRDS(file = here::here("data", "gss2024.rds"))

gss <- gss2024 |>
  select(courtsnv) |>
  drop_na() |>
  haven::zap_labels()

gss_recoded <- gss |>
  mutate(
    courtsdv = recode(courtsnv,
                      `1` = -1,
                      `2` = 1,
                      `3` = 0)
    )

gss_recoded <- gss_recoded |>
  mutate(dev = courtsdv - 0)
```

```r
mod_c <- lm(dev ~ 0,
            data = gss_recoded)

mod_a <- lm(dev ~ 1,
            data = gss_recoded)
df_1 <- 1
df_2 <- 615
```

### 5b. Report PRE, F*, and t*

```r
sse_mod_c <- deviance(mod_c)
sse_mod_a <- deviance(mod_a)

pre_5 <- (sse_mod_c - sse_mod_a) / sse_mod_c
pre_5
```

```
[1] 0.2939893
```

```r
n_5 <- gss_recoded |>
  nrow()

f_star <- (pre_5 / (pa - pc)) / ((1 - pre_5) / ( n_5 - 1))
f_star
```

```
[1] 256.0916
```

```r
t_star <-sqrt(f_star)
```

### 5c. Use tables or functions to identify critical values and write conclusions

```r
# PRE
calc_null_pre_5 <- function() {

  null_data_5 <- tibble(dev = rnorm(615, 0, 0))

  m_0 <- lm(dev ~ 0, data = gss_recoded)
```

```r
  m_1 <- lm(dev ~ 1, data = gss_recoded)


  null_pre <- (deviance(m_0) - deviance(m_1)) / deviance(m_0)

  return(null_pre)
}

null_sims_5 <- tibble(sim_number = 1:1000) |>
  rowwise() |>
  mutate(pre = calc_null_pre_5())

quantile(null_sims$pre, c(.95, .99, 1))
```

```
      95%       99%      100%
0.1177732 0.2035586 0.3673728
```

```r
# F
f_c <- qf(1 - alpha, df_1, df_2)
f_c
```

```
[1] 3.856624
```

```r
# t
f_t <- qt(1 - alpha, df_2, lower.tail = TRUE)
f_t
```

```
[1] 1.647335
```

I reject the null hypothesis because the observed F statistic of 256.092 exceeds the F* of 3.857. This means that respondents' answers are not evenly distributed between "too harshly" and "not harshly enough"; the mean is not 0.